

4.2 Power Series

In the last section we looked n th Maclaurin and Taylor polynomials which were used to approximate a function f about a point $x = x_0$. It seemed reasonable that as n increased $p_n(x)$ would provide better and better approximations, and may even possible converge to $f(x)$ as $n \rightarrow \infty$.

Definition: If f has derivatives of all orders at x_0 , then we call the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!} (x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n + \dots$$

the **Taylor series for f about $x = x_0$** .

In the special case where $x = 0$, this series becomes the **Maclaurin series for f**

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \dots + \frac{f^{(n)}(0)}{n!} x^n + \dots$$

Maclaurin and Taylor series differ from the series that we considered in 3.2 – 3.8 in that their terms are not merely constants, but also involve variables. These are examples of what we call a *power series*.

Definition: A power series is a series of the form $\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$ where x is a variable and the c_n 's are constants called the coefficients of the series.

Note: We can think of a power series as a function of x . The domain of a power series is the values of x for which the series converges.

Note: Intuitively, one might think of a power series as a polynomial with infinite degree.

Example 1: Consider the power series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$.

For this power series $c_n = 1$ for all $n \geq 0$. Moreover, by our findings with geometric series, we know that this series converges when $|x| < 1$ and diverges when $|x| \geq 1$.

Note: If we view $\sum_{n=0}^{\infty} x^n$ as function of x . For example, $f(x) = \sum_{n=0}^{\infty} x^n$. Then, the domain of this function is $(-1, 1)$.

Definition: A series of the form $\sum_{n=0}^{\infty} c_n (x-x_0)^n = c_0 + c_1(x-x_0) + c_2(x-x_0)^2 + \cdots$ is called a power series in $(x-x_0)$ or a power series centered at x_0 or a power series about x_0 .

Note: As is indicated by the above definition for power series, we adopt the convention that $(x-x_0)^0 = 1$ even when $x = x_0$.

Note: When $x = x_0$ most all the terms become 0 and the above power series always converges when $x = x_0$.

Example 2: For what values of x is the series $\sum_{n=0}^{\infty} n!x^n$ convergent?

For this question we will use the ratio test. Using our typical series notation, in this example, $a_n = n!x^n$. Now,

we calculate that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)!x^{n+1}}{n!x^n} \right| = \lim_{n \rightarrow \infty} |(n+1)x| = \lim_{n \rightarrow \infty} (n+1)|x|$. We note that if $x \neq 0$, then

$\lim_{n \rightarrow \infty} (n+1)|x| = \infty$, and if $x = 0$, then $\lim_{n \rightarrow \infty} (n+1)|x| = 0$. So, by the ratio test the given series converges only when $x = 0$.

Example 3 (you try): For what values of x does the series $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converge?

For this question we will use the ratio test. Using our typical series notation, in this example, $a_n = \frac{(x-3)^n}{n}$.

Now, we calculate that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\left(\frac{(x-3)^{n+1}}{n+1} \right)}{\left(\frac{(x-3)^n}{n} \right)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-3)n}{n+1} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \cdot \frac{1/n}{1/n} \right) |x-3| = \lim_{n \rightarrow \infty} \left(\frac{1}{1+1/n} \right) |x-3| = |x-3|.$$

So, by the

ratio test we know that this series is convergent when $|x-3| < 1 \Leftrightarrow 2 < x < 4$, and it is divergent when $|x-3| > 1$.

The ratio test is inconclusive when $x = 4$ and $x = 2$. So, we will analyze these cases separately. When $x = 4$

the series becomes $\sum_{n=1}^{\infty} \frac{(4-3)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$, which is the harmonic series. We know the harmonic series is divergent,

which means that the series is divergent when $x = 4$. When $x = 2$ the series becomes $\sum_{n=1}^{\infty} \frac{(2-3)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$,

which we know converges by the Alternating Series Test. Thus, the power series converges for $2 \leq x < 4$.

Theorem: For a given power series $\sum_{n=0}^{\infty} c_n (x - x_0)^n$ there are only three possibilities:

1. The series converges only when $x = x_0$.
2. The series converges for all x .
3. There is a positive number R such that the series converges if $|x - x_0| < R$ and diverges if $|x - x_0| > R$.

Proof: Beyond the scope of the course.

Definition: The interval of convergence of a power series is the interval that consists of all values of x for which the series converges.

Definition: The number R in case three of the theorem is called the radius of convergence of the power series. By convention we let $R = 0$ in case one of the theorem and we let $R = \infty$ in case two of the theorem.

Note: In case 1 of the theorem the interval of convergence consists of a single point a . In case 2 of the theorem the interval of convergence is $(-\infty, \infty)$. In case 3 of the theorem the interval of convergence is $(x_0 - R, x_0 + R)$, $(x_0 - R, x_0 + R]$, $[x_0 - R, x_0 + R)$, or $[x_0 - R, x_0 + R]$.

Example 4: Consider the series (from the previous example): $\sum_{n=0}^{\infty} n! x^n$.

The radius of convergence of this power series is $R = 0$. The interval of convergence for this power series is $[0, 0]$.

Example 5: Consider the series (from the previous example): $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$.

The radius of convergence of this power series is $R = 1$. The interval of convergence for this power series is $[2, 4)$.

Note: If we view $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ as function of x . For example, $f(x) = \sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$. Then, the domain of this function is $[2, 4)$.

Example 6 (you try): Find (with justification) the radius of convergence and interval of convergence of the

$$\text{series: } \sum_{n=1}^{\infty} \left(\frac{(2x-1)^n}{5^n \sqrt{n}} \right).$$

For this question we will use the ratio test. Now, we calculate that

$$\lim_{n \rightarrow \infty} \left| \frac{\left(\frac{(2x-1)^{n+1}}{5^{n+1} \sqrt{n+1}} \right)}{\left(\frac{(2x-1)^n}{5^n \sqrt{n}} \right)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(2x-1) \sqrt{n}}{5 \sqrt{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{1}{5} \left(\sqrt{\frac{n}{n+1}} \cdot \frac{1/n}{1/n} \right) |2x-1| = \frac{1}{5} \left(\sqrt{\lim_{n \rightarrow \infty} \frac{1}{1+1/n}} \right) |2x-1| = \frac{1}{5} |2x-1|. \text{ So, by the}$$

ratio test we know that this series is convergent when $\frac{1}{5} |2x-1| < 1 \Leftrightarrow -5 < 2x-1 < 5 \Leftrightarrow -4 < 2x < 6 \Leftrightarrow -2 < x < 3$, and it is divergent when $\frac{1}{5} |2x-1| > 1$. The ratio test is inconclusive when $x = -2$ and $x = 3$. So, we will analyze these cases separately.

When $x = -2$ the series becomes $\sum_{n=1}^{\infty} \left(\frac{(2(-2)-1)^n}{5^n \sqrt{n}} \right) = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$. This is an alternating series with $a_n = \frac{1}{\sqrt{n}}$.

We note that for $n \geq 1$ $\sqrt{n+1} > \sqrt{n} \Leftrightarrow \frac{1}{\sqrt{n+1}} < \frac{1}{\sqrt{n}} \Leftrightarrow a_{n+1} < a_n$. Furthermore, $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = \sqrt{\lim_{n \rightarrow \infty} \frac{1}{n}} = 0$.

So, this series converges by the Alternating Series Test.

When $x = 3$ the series becomes $\sum_{n=1}^{\infty} \left(\frac{(2(3)-1)^n}{5^n \sqrt{n}} \right) = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$, which we know diverges because it is a p -series with p value of one half.

These facts imply that the radius of convergence is $R = 2.5$ and the interval of convergence is $[-2, 3)$.